

Verification Test Cases

Regression Testing Matrix & Validation SOPs for Real-Time Detection

1. Introduction

This test matrix provides step-by-step verification protocols to guarantee the mathematical accuracy, role security enforcement, and reporting integrity of the **Anomaly.Stream** system.

2. Functional Test Cases

TC-001: Autonomic Sliding Z-Score Validation

* **Preconditions:** Dynamic stream is `LIVE`. Ingestion interval set to `1000ms`.

* **Action:**

1. Open the System Dashboard tab.
2. Locate the **Injections Panel** under **Interactive Telemetry Stress Injections**.
3. Select metric `API Latency` and request a `SPIKE` with a multiplier of `15.0x`.

* **Expected Result:**

- * System immediately registers a metric event with score value exceeding standard boundary.
- * A sound-alarm or visual banner is thrown instantly.
- * A new anomaly ticket is created in the Anomalies Registry with status `Active` and Severity `Critical` or `High`.

TC-002: Role Privilege Boundary Integrity (Viewers Lockout)

* **Preconditions:** Authenticated under email `viewer@anomaly.io` (assigned role **Viewer**).

* **Action:**

1. Navigate to the **Settings** view.
2. Attempt to drag the "Anomaly Z-Score Threshold" slider or input a new value.
3. Attempt to click the "Hard Purge Historical logs" button.

* **Expected Result:**

- * Control sliders are disabled visually (`disabled` attribute activated).
- * Attempting to write modifications results in security error box: **"Security Policy Error: Viewers do not have roles permission to modify system constants."**
- * Admin purge buttons are hidden or fully blocked.

TC-003: Gemini Diagnosis Report Synthesis

* **Preconditions:** At least 3 active anomalies registered. Gemini API key configured on server.

* **Action:**

1. Navigate to **AI Analytics**.
2. Click `COMPILE SYSTEM AI REPORT`.

* **Expected Result:***

- * "Consulting Neural Cores..." load state triggers.
- * Client sends system stats, recent anomalies, and stream contexts in the request payload.
- * Output area streams or renders a cleanly formatted markdown summary containing:
 - Overall Threat Summary
 - Multi-Thread Correlative Analysis
 - Remediation SOP Recommendation.